

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

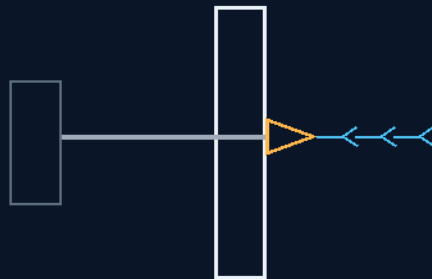
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 187

THE QUINN THE BACKDRAFT BAYONET

Breach, flood, look, plug
the door stays shut



the M60 rule — crew-served by design

25MM BAYONET · TWO BOTTLES · TWO TRIGGERS

NO OXYGEN GIFT

Fire system — v1.0 in force

GP-IM-187

Revision v1.0 — 23 August 2026

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VETTED BATCH 23 — PASS · WORDING RIDER CARRIED

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 187

PHASE 187: THE QUINN THE BACKDRAFT BAYONET — THROUGH-DOOR CO2 ATTACK (THE M60 RULE)
— STATUS: CURRENT — PASS; THE VET'S WORDING RIDER CARRIED — ABSOLUTE-SAFETY LANGUAGE REMAINS AN OBJECTIVE, NOT A GUARANTEE. Earth fire safety, and the scenario every firefighter knows: searching rooms, the door is hot — you leave it, because a hot door marks a compartment waiting to backdraft, and opening it feeds the fire the oxygen it is starving for. The existing options are all compromises: wait for water and ventilation coordination, or take the gamble. Phase 187 attacks through the door. M60-sized, crew-served, two bottles and two triggers: compressed air drives a fixed hollow bayonet — 25mm, 10mm bore, pointed cone with holes, one ring — through the door; the second trigger floods the compartment with CO2 through the breach, no doorway opening, no oxygen gift. Look, then plug: a short cable camera to a screen checks through the breach whether flame persists; if entry stays non-viable, the number-2's cone plug seals the hole. Crew-served by design — in battle the man with the M60 has a number-2 gunner who carries all the crap: spare bottles, cone plugs, the camera. The vet passed it with the wording rider carried: absolute-safety language remains an objective, not a guarantee.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-187, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the bayonet law as seated (contents line, the bodies — the problem, the weapon, the number-2 doctrine — the audit and provenance, verbatim) — §2 why through the door (graded) — §3 the system through the door — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 187: **The Backdraft Bayonet — Through-Door CO2 Attack (The**

M60 Rule). M60-sized two-bottle fire weapon for the hot-door gamble: compressed air drives a fixed hollow bayonet (25mm, 10mm bore, cone point, discharge ring 10mm behind the tip) through the door, and the second trigger immediately floods the compartment with CO2 directly behind the door — attack without feeding air. Cable-cam check through the breach; red silicon cone plug seals the hole, no drilling; walls breached the same way. Crew-served: the number-2 gunner carries spare bottles, plugs and cam.*'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of the Hot-Door Gamble — attack the fire without feeding it air '

Fire Suppression Baseline: Through-Door Inerting for Compartment Fires (Earth Scope)

Live instruction, 30 July 2026. Earth fire safety, first build of the sitting. The scenario every firefighter knows: searching rooms in a burning building, the door is hot, you leave it — because opening it invites backdraft. Archer's answer: don't open the door. Punch a hollow bayonet through it, flood the compartment with CO2 directly behind the door, look, and plug. Crew-served, M60-style: the gunner fights, the number-2 carries the rest.

THE PROBLEM — THE HOT DOOR GAMBLE (VERBATIM)

- **The scenario, in Archer's words:** firefighters go into a building searching rooms; the door is hot; you leave it — backdraft. A hot door marks a compartment full of superheated, oxygen-starved fuel gases; open it and the incoming air can ignite the whole volume at once, with the crew standing in the doorway. The choice crews hate: leave the room to burn, or gamble the door.
- **The existing options are all compromises:** wait for water and ventilation coordination, or take the gamble. Nothing in the standard kit attacks the compartment while keeping the door shut and the air out.

THE WEAPON — TWO BOTTLES, TWO TRIGGERS (VERBATIM)

- **The loadout:** two cylinders — one a normal CO2 extinguisher bottle, standard small car size; the other compressed air, the driver. M60-sized, heavy like a ram: the weight is a feature, making the gun a stable firing platform quickly placed against the door.
- **The bayonet (his spec, verbatim dimensions):** fixed bayonet, 25mm diameter, hollow with a 10mm bore; the end is a pointed cone with holes — one ring of holes on the straight section, 10mm back from the end of the cone point. The cone does the piercing; the ring sits just past the inner face so discharge begins inside the compartment, not inside the door's thickness.
- **The sequence:** placed against the door and fired — compressed air drives the bayonet through — and immediately the operator presses the second trigger, flooding the room with CO2 through the hollow bayonet, importantly directly behind the door. No oxygen window: the same breach that

delivers the gas never opens the room to air. Wall breaches work the same.

- **Look, then plug:** a short cable camera to a screen checks through the breach whether flame persists. If entry remains non-viable, the number-2's red silicon cone plug seals the 25mm hole — no drilling — and the inert atmosphere is left inside to finish the work while the crew moves on.

THE NUMBER-2 GUNNER DOCTRINE (VERBATIM)

- **Crew-served by design:** in battle, the man with the M60 has a number-2 gunner who carries all the crap — spare bottles, cone plugs, the camera, everything that isn't the gun. The weapon is heavy because it must be stable; the team exists so the weight never slows the attack. Reload and re-engage is a drill, not an improvisation.

PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED WITH FLAGS — THE PREMISE IS TEXTBOOK, THE DELTAS ARE REAL, AND THE VOLUME MATH MUST BE SAID PLAINLY) (VERBATIM)

The premise is correct fire science: a sealed compartment fire burns down its oxygen, fills with hot pyrolysis gases, and sits one oxygen-rich breath away from backdraft — attacking without opening is the right doctrine, and fire services already practice indirect attack for exactly this reason. Prior art, recorded per the file's recognition doctrine: piercing nozzles and fog-spike devices that are hammered or driven through walls and then flow WATER are established equipment — Archer's deltas are (1) compressed-air drive, so there is no hammer swing and no repeated striking in a smoke-filled corridor; (2) CO₂ inerting instead of water — no water damage, safe on electrical and fuel loads, and it starves the fire rather than only cooling it; (3) the discharge ring positioned to begin directly behind the door, inerting the entry zone first; and (4) the plug-and-leave seal that keeps the inert atmosphere in the compartment. The flags, kept honest: (1) VOLUME MATH — a car-size ~2kg CO₂ bottle yields roughly one cubic metre of gas; that floods and knocks down the doorway zone and buys real time, but it will not inert a whole 30-cubic-metre room in a single shot. The number-2's spare bottles are the volume answer, matched to compartment size — the doctrine already carries them. (2) Drive energy depends on door material — timber and composite doors, yes; steel fire doors are a power question for the bench. (3) The hole ring must clear the inner face of the thickest expected door — a tuning spec, his 10mm dimension recorded as given. (4) The CO₂/breathing law of Phases 163/178 applies: in a fully involved compartment, any casualty is already breathing smoke; occupied-building protocol is a fire-service SOP decision, flagged for the crews, not invented by the file. None of the flags close the door on the build.

ORIGIN OF THOUGHT — PHASE 187 (VERBATIM)

Archer, 30 July 2026: "ok fire safety earth, the m60 sized anti fire weapon. firefighters go into a building searching rooms, door is hot, you leave it, backdraft. but this is a game changer, 2 cylinders one is a normal co₂ extinguisher bottle, standard small size for a car. the other is compressed air, the barrel is

placed against the door to fire a fixed bayonet through the door, the bayonet is 25mm diameter the end is a pointed cone with holes, 1 ring of holes on the straight section 10mm from the end of the cone point, the bayonet is hollow 10mm bore. the heavy weight gun like a ram is quickly placed against the door, fired and immediately the operator presses the second trigger, flooding the room with CO₂, and importantly directly behind the door, wall breaches could work the same. short cable cam to screen to see if there is still flame, not viable, the other guy has the pod room red silicon cone plug, no need to drill. in battle the guy with the M60 has a number 2 gunner who carries all the crap. spare bottles etc, i can see the FBI and police saying ooooh tear gas"

Dual-use ledger, recorded as he said it: his own closing observation — the FBI and police will look at a through-door gas injector and think tear gas. The file records this openly rather than pretending otherwise: any through-door delivery system is inherently dual-use, and the CERN-OHL publication means the design is free for fire services and visible to everyone else at the same time. His stated intent and scope is fire suppression, Earth; the ledger entry exists so the record is honest.

Why it matters, in plain English: the hot-door gamble kills the people who came to help. This turns the gamble into a procedure — punch, flood, look, plug — and the procedure is crew-served, so no one is a lone hero with a heavy tool in a corridor. Note the fingerprint: a proven concept from one domain (the M60's number-2 gunner) transplanted to a new one, the same instinct that put a wiper motor on a gasifier and kids' shoe lights on a battlestar. He doesn't invent parts; he re-assigns proven ones.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PREMISE IS TEXTBOOK, AFFIRMED: a hot door marks a ventilation-limited compartment fire — rich in unburned fuel, starved of oxygen; opening the door is the classic backdraft trigger. Attacking the compartment without opening the path is the correct tactical answer, and the standard kit has no clean tool for it.

THE MECHANISM IS PROVEN PARTS: pneumatic piercing and CO₂ flooding are both established — the bayonet is a driven hollow spike, the flood is a standard extinguisher bottle. The invention is the sequence: breach, flood, look, plug — through one 25mm hole.

THE LOOK-THEN-PLUG, AFFIRMED AS DOCTRINE: the camera check converts a blind gamble into an instrumented decision; the cone plug converts a failed entry into a sealed compartment. Every state of the fight has a next move.

THE HONEST FLAGS (the audit's, seated): the deltas are real — door construction varies (steel-clad, solid-core, fire-rated), penetration is not guaranteed against every door, and CO₂ flooding a compartment that may hold a casualty is a triage decision, not a default. The vet's wording rider is carried: the weapon makes the gamble smaller, never zero.

§3 — THE SYSTEM THROUGH THE DOOR

- THE LOADOUT: two cylinders — CO2 extinguisher bottle plus compressed-air driver; M60-sized, heavy, crew-served.
- THE BAYONET: fixed, 25mm, 10mm bore, pointed cone with holes — his spec, verbatim dimensions.
- THE SEQUENCE: placed against the door, fired — air drives the bayonet through; second trigger floods CO2.
- THE LOOK: cable camera through the breach — flame check on the screen.
- THE PLUG: the number-2's cone plug seals the hole if entry stays non-viable.
- THE NUMBER-2: spare bottles, plugs, camera — the carrier who makes the weapon a system.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE NO-OXYGEN-GIFT DOCTRINE: attack the compartment, never feed it — the door stays shut.
- THE CREW-SERVED DOCTRINE: heavy weapons carry a number-2 — the M60 rule applied to fire.
- THE LOOK-THEN-PLUG DOCTRINE: instrument the breach; seal what you cannot enter.
- THE HONEST-GAMBLE DOCTRINE: the weapon shrinks the gamble; the file says it never removes it.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the bayonet: 25mm hollow spike, 10mm bore, cone-hole pattern — penetration per door class benched.
- STEP 2 — Bench the drive: air pressure vs penetration — the honest door-class envelope.
- STEP 3 — Bench the flood: CO2 discharge through the breach — compartment knockdown timing.
- STEP 4 — Bench the look and plug: camera through the bore, cone-plug seal under heat.
- STEP 5 — Drill the crew: two-person sequence, number-2 doctrine — then publish, open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 178 — the suppression disc (GP-IM-178): the corridor companion — the disc slides, the bayonet breaches.

- PHASE 152 — the blast-dampening valves (GP-IM-152): pressure-event kin.
- PHASE 163 — the casualty pods (GP-IM-163): the triage doctrine the flood decision rides.
- PHASE 173 — the fire law (GP-IM-173): elimination first; this is for the fires that happen anyway.

§8 — DO-NOT-BUILD

- DO NOT open the hot door — the bayonet exists so the door stays shut.
- DO NOT promise every door — construction varies; the penetration envelope is benched and published.
- DO NOT flood blind — look first; a compartment that may hold a casualty is a triage decision.
- DO NOT call it safe — the vet's rider is law: safety is the objective, never the guarantee.

§9 — OWED

OWED: the door-class penetration envelope and compartment knockdown timing (bench, §6); the vet's wording rider is carried inline in §1 and §8.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-187, v1.0 — CURRENT — PASS; the vet's wording rider carried — absolute-safety language remains an objective, not a guarantee, seated law, master v2.1 (numerical print order). Built 23 August 2026, twentieth of the 20-chapter 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 187